

Boris Khesin

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EDUCATION

- **Grinnell College** Grinnell, Iowa
BA in Economics & Computer Science; GPA: 3.78 August 2024 – May 2028 (expected)
 - **Relevant Coursework:** Computer Organization & Architecture (CSC-211), Econometrics (ECN-286), International Economics (ECN-233), Applied Statistics (STA-209), Functional Programming (CSC-151), Microeconomic Analysis (ECN-280), Introduction to Economics (ECN-111)
- **Moscow State School 57** Moscow, Russia
Mathematics & Computer Science track; GPA: 4.0 September 2020 – June 2024

RESEARCH

- **State Corporate Income Taxes and New Business Formation in the United States** May 2026
Course research paper (with B. Pilling), Grinnell College
 - **Overview:** Econometric study of whether top marginal state corporate income tax rates deter new firm entry, using a balanced state-year panel (51 jurisdictions, 2004–2020; 867 observations) built from Census Business Formation Statistics, Tax Foundation rate series, and BEA regional accounts. Estimated pooled OLS, two-way fixed-effects, and 2SLS specifications in Stata with state-clustered standard errors, instrumenting with regional-average tax rates; ran robustness checks including large-state exclusion and a one-year-lead placebo test. The negative cross-sectional association does not survive fixed-effects or IV identification.
- **Economic, Demographic, and Policy Factors in COVID-19 Mortality**
 - **Across Countries**
Course research paper (with B. Pilling and J. Pada), Grinnell College December 2025
 - **Overview:** Cross-national analysis of COVID-19 deaths per capita and per case across 150 countries using Jiangsu University, CIA World Factbook, and World Bank data. Fit univariable and multivariable linear regressions in R with nested-model F-tests, log transformations, and residual diagnostics; identified HDI as the strongest predictor of deaths per capita and population density of deaths per case, interpreting both through institutional capacity and measurement quality.

WORK EXPERIENCE

- **Topcon Consulting (client: Watson-Forsberg)** Minneapolis, Minnesota
Technology Consulting Intern June 2026 – July 2026
 - **IT Support:** Served as first-line technology contact for office and construction staff at Watson-Forsberg, a Minneapolis construction firm, resolving access and identity questions across Unanet, Procore, and Microsoft Entra and advising end users on the firm's core enterprise platforms.
 - **Data Analysis:** Analyzed event spending across 50+ company events involving hundreds of guests and dozens of clients; consolidated the figures into budget projections and cost breakdowns that informed planning and vendor decisions.
 - **Software Development:** Scoped and built a custom internal event-planning application in Django after gathering requirements directly from end users, integrating it with Unanet, Procore, and Microsoft systems and adding budget projections and Google Maps.
- **Grinnell College Economics Department** Grinnell, Iowa
Teaching Assistant, International Economics (ECN-233) January 2026 – May 2026
 - **Economics Instruction:** Guided 20+ students through trade theory, international finance, and economic policy; led weekly review sessions analyzing exchange rates, trade agreements, and globalization with real-world data, and gave analytical feedback on quantitative problem sets.
- **Grinnell College Statistics Department** Grinnell, Iowa
Teaching Assistant, Applied Statistics (STA-209) January 2026 – May 2026
 - **Quantitative Methods:** Led weekly sessions on regression, hypothesis testing, and probability using R and Excel, translating technical concepts for students across a range of quantitative backgrounds.

- **Grinnell College Computer Science Department** Grinnell, Iowa
Grader, Functional Programming (CSC-151) September 2025 – December 2025
 - **Evaluation & Feedback:** Evaluated assignments and exams for a Scheme/Racket functional programming course, writing feedback on code correctness and algorithm design; handled grades with strict confidentiality and managed grading workflows independently to weekly deadlines.
- **Grinnell College Swimming Team** Grinnell, Iowa
Event Worker December 2024 – May 2025
 - **Lane Timer:** Recorded backup times and verified results at competitive meets, including official NCAA events; documented race results to support fair competition and official scoring.
- **Yandex** Moscow, Russia
Mentor/Tutor, Yandex Academy December 2023 – August 2024
 - **Technical Mentoring:** Tutored students in C++ and Python one-on-one and in small groups of 3–5; reinforced core programming concepts and supported students' Django projects through code review and step-by-step instruction.
- **Tinkoff Bank** Moscow, Russia
Product Manager Intern April 2024 – May 2024
 - **Central University App:** Co-led a development team building a React.js Q&A and gamification app for Central University students; created Figma prototypes and user flows and coordinated sprint work with the development team in Trello.

PROJECTS

- **PIPS — 16-bit CPU & Assembler:** Designed and implemented a microcoded 16-bit CPU in Logisim with a partner for Computer Organization & Architecture (CSC-211): built the datapath with custom ALU, register-file, and shifter subcircuits; wrote the control-ROM microprogram driving 16 opcodes; defined Python assembler rules for 40+ instructions and pseudo-instructions; verified the processor with 14 assembly test programs spanning recursion, arithmetic, and memory-mapped keyboard/terminal I/O.
- **Event Planner (Django):** Built an internal event-planning web app for Watson-Forsberg; synchronized with Unanet, Procure, and Microsoft corporate systems, with budget projections and Google Maps integration. (github.com/Shagulka/Event-Planner)
- **Tarot App (iOS):** Built an iOS app for AI-powered tarot readings in Swift and SwiftUI; integrated LLM-generated interpretations for multiple card spreads and designed the interface to Apple Human Interface Guidelines. (github.com/Shagulka/Tarot-App)
- **Non-Human Vision (VR/XR):** Created a VR/XR simulation of non-human animal visual systems in Unity and C# with a team, writing custom shaders to replicate species-specific color perception and fields of view.

LEADERSHIP & AWARDS

- **Debate Leader, Grinnell College Debate Union (Fall 2025 – Present):** Direct weekly meetings, organize travel and resources, and mentor members on argumentation and debate strategy.
- **Second Place, Pioneer Weekend Entrepreneurial Contest:** Led a team of 4 designing a campus vending-machine business with ROI projections and data analytics; earned second place and personal recognition from Grinnell College President Anne Harris.
- **AI & Competitive Programming, Centre for Teaching Excellence (2023):** Self-taught transformer architectures in three weeks and fine-tuned a GPT language model to imitate friends' group-chat texting style, engineering my own compute access on limited hardware; presented the work at an AI conference, which led to a product internship at Tinkoff Bank. Also trained in competitive programming: automata, MST, dynamic programming, graphs, strings, and segment trees.

SKILLS

- **Quantitative & Analytical:** R, Stata, Python (pandas, NumPy), Excel; regression, hypothesis testing, econometric analysis (panel data, fixed effects, instrumental variables), data cleaning and visualization
- **Programming Languages:** Python, C++, Swift, R, SQL, JavaScript, HTML/CSS, Scheme/Racket
- **Frameworks & Tools:** Django, React.js, SwiftUI, Unity (C#), WebGL; Git, LaTeX, Logisim, Figma, Trello; Unanet, Procure, Microsoft Entra
- **Languages:** English (fluent), Russian (native)